

IoT protocols at link layer

Zigbee. Zigbee is an IEEE 802.15.4-based specification for a suite of high-level communication protocols used to create personal area networks with small, low-power digital radios, such as for home automation, medical device data collection, and other low-power low-bandwidth needs, designed for small scale projects which need wireless connection. Hence, Zigbee is a low-power, low-data-rate, and close proximity (i.e., personal area) wireless ad hoc network.

The technology defined by the Zigbee specification is intended to be simpler and less expensive than other wireless personal area networks (WPANs), such as Bluetooth or more general wireless networking such as Wi-Fi (or Li-Fi). Applications include wireless light switches, home energy monitors, traffic management systems, and other consumer and industrial equipment that requires short-range low-rate wireless data transfer.

Its low power consumption limits transmission distances to 10–100 meters (33–328 ft) line-of-sight, depending on power output and environmental characteristics. Zigbee devices can transmit data over long distances by passing data through a mesh network of intermediate devices to reach more distant ones. Zigbee is typically used in low data rate applications that require long battery life and secure networking. (Zigbee networks are secured by 128-bit symmetric encryption keys.) Zigbee has a defined rate of up to 250 kbit/s, best suited for intermittent data transmissions from a sensor or input device.

Bluetooth. Bluetooth is a short-range wireless technology standard that is used for exchanging data between fixed and mobile devices over short distances and building personal area networks (PANs). In the most widely used mode, transmission power is limited to 2.5 milliwatts, giving it a very short range of up to 10 metres (33 ft). It employs UHF radio waves in the ISM bands, from 2.402 GHz to 2.48 GHz. It is mainly used as an alternative to wired connections to exchange files between nearby portable devices and connect cell phones and music players with wireless headphones, wireless speakers, HIFI systems, car audio and wireless transmission between TVs and soundbars.

Bluetooth Low Energy (BLE), a form of Bluetooth optimized for low-powered devices, can assist IoT devices in conserving energy by keeping them in sleep mode until they are connected. BLE is perfect for IoT applications since it can pair and reconnect with devices in six milliseconds instead of six seconds with traditional Bluetooth. This not only improves operating efficiency but also increases device availability.

The hardware, communication, software system, and application layers make up a typical IoT design, with Bluetooth serving as the communication layer. The communication layer, which consists of a multi-layer stack that includes data connection, network or transport, and session protocols, is a vital link between the layers. Bluetooth, often known as BLE, is a data communication layer that links sensors to sensors or sensors to gateways. On the other hand, the network layer is in charge of routing or transporting packets across the network using the best pathways possible. The session layer protocols allow communication between different aspects of the IoT communication subsystem.

Bluetooth technology is less expensive to implement than other communication protocols such as RFID, NFC, WLAN, LoRa WAN, LTE-A, and WiFi-Direct. It also allows for wireless communication and the creation of an immediate Personal Area Network (PAN) in areas where wireless infrastructure is lacking. It features a uniform methodology and minimum interference. Furthermore, by increasing the power beyond one milliwatt, Bluetooth's standard range of 0-30 meters may extend.

Wi-Fi. Wi-Fi is a family of wireless network protocols based on the IEEE 802.11 family of standards, which are commonly used for local area networking of devices and Internet access, allowing nearby

digital devices to exchange data by radio waves. These are the most widely used computer networks, used globally in home and small office networks to link devices and to provide Internet access with wireless routers and wireless access points in public places such as coffee shops, restaurants, hotels, libraries, and airports.

Wi-Fi uses multiple parts of the IEEE 802 protocol family and is designed to work well with its wired sibling, Ethernet. Compatible devices can network through wireless access points with each other as well as with wired devices and the Internet. Different versions of Wi-Fi are specified by various IEEE 802.11 protocol standards, with different radio technologies determining radio bands, maximum ranges, and speeds that may be achieved. Wi-Fi most commonly uses the 2.4 gigahertz (120 mm) UHF and 5 gigahertz (60 mm) SHF radio bands, with the 6 gigahertz SHF band used in newer generations of the standard; these bands are subdivided into multiple channels. Channels can be shared between networks, but, within range, only one transmitter can transmit on a channel at a time.

Wi-Fi's radio bands work best for line-of-sight use. Common obstructions, such as walls, pillars, home appliances, etc., may greatly reduce range, but this also helps minimize interference between different networks in crowded environments. The range of an access point is about 20 m (66 ft) indoors, while some access points claim up to a 150 m (490 ft) range outdoors. Hotspot coverage can be as small as a single room with walls that block radio waves or as large as many square kilometers using multiple overlapping access points with roaming permitted between them. Over time, the speed and spectral efficiency of Wi-Fi have increased. As of 2019, some versions of Wi-Fi, running on suitable hardware at close range, can achieve speeds of 9.6 Gbit/s (gigabit per second).

IoT protocols at network layer

RPL. RPL (Routing Protocol for Low-Power and Lossy Networks) is a routing protocol for wireless networks with low power consumption and generally susceptible to packet loss. It is a proactive protocol based on distance vectors and operates on IEEE 802.15.4, optimized for multi-hop and many-to-one communication, but also supports one-to-one messages.

This protocol is specified in RFC 6550 with special applications in RFCs 5867, 5826, 5673 and 5548. RPL can support a wide variety of link layers, including those with limitations, with potential losses or that are used in devices with limited resources. This protocol can quickly create network routes, share routing knowledge and adapt the topology in an efficient way.

RPL creates a topology similar to a tree (DAG or directed acyclic graph). Each node within the network has an assigned rank (Rank), which increases as the teams move away from the root node (DODAG). The nodes resend packets using the lowest rank as the route selection criteria.

The implementation of the RPL protocol occurs in wireless sensors and networks, the most used operating system for its implementation is Contiki which is a small open source operating system developed for use in a number of small systems ranging from 8-bit computers to integrated systems on microcontrollers, including sensor network nodes.

CORPL. CORPL protocol is the extension of the RPL protocol, which is termed as cognitive RPL. This network protocol is designed for cognitive networks and uses DODAG topology. CORPL protocol makes two new modifications in the RPL protocol. It uses opportunistic forwarding to forward a packet between the nodes. Each node of CORPL protocol keeps the information of the forwarding set rather than parents only maintaining it. Each node updates its changes to its neighbor using DIO messages. On the basis of this updated message, each node frequently updates its neighbor for a constant forwarder set.

CARP. CARP (Channel-Aware Routing Protocol) is a distributed routing protocol. It is designed for

underwater communication. It has lightweight packets so that it can be used for Internet of Things (IoT). It performs two different functionalities: network initialization and data forwarding. CARP protocol does not support previously collected data. Hence, it is not beneficial for those IoT or other applications where data is changed frequently. The upgradation of CARP is done in E-CARP which overcomes the limitation of CARP. The E-CARP allows the sink node to save previously received sensory data.

6LoWPAN. 6LoWPAN stands for IPv6 over Low-power Wireless Personal Area Networks. It is a standard protocol for realizing IPv6 communication on wireless networks composed of low-power wireless modules. 6LoWPAN specification contains packet compression and other optimization mechanisms to enable the efficient transmission of IPv6 packets on a network with limited power resources and reliability, which makes efficient IPv6 communication over low-power wireless networks possible.

The 6LoWPAN group defined encapsulation, header compression, neighbor discovery and other mechanisms that allow IPv6 to operate over IEEE 802.15.4 based networks. Although IPv4 and IPv6 protocols do not generally care about the physical and MAC layers they operate over, the low-power devices and small packet size defined by IEEE 802.15.4 make it desirable to adapt to these layers.

Various low-power wireless networks have been proposed and implemented before 6LoWPAN, but currently, 6LoWPAN is regarded as one of the preferred protocols to realize the Internet of Things (IoT). This is because 6LoWPAN communication based on IPv6 allows once closed low-power wireless networks to interface with the global network, the Internet, and implement more advanced intelligent services that were not possible in the past.

IoT protocols at transport layer

TCP. TCP is a connection-oriented protocol that ensures reliable and ordered delivery of data. It is widely used in traditional networking and provides mechanisms like acknowledgments, retransmissions, and congestion control. This makes TCP well-suited for IoT applications where data integrity is critical, such as firmware updates or secure communication. However, it comes with added overhead due to its connection setup (three-way handshake) and stateful nature, which can be problematic for low-power or memory-constrained IoT devices. Additionally, its heavier structure can cause delays, making it less ideal for time-sensitive communication.

UDP. UDP is a connectionless protocol known for its low latency and minimal overhead. It does not provide guarantees for delivery, ordering, or duplication, making it significantly lighter than TCP. UDP is ideal for real-time applications like voice, video, or sensor data streaming, where speed is more critical than reliability. However, because UDP lacks built-in mechanisms for reliability, any necessary error-checking or retransmission logic must be implemented at the application layer.

MQTT. Built on top of TCP, MQTT (Message Queuing Telemetry Transport) is a lightweight publish/subscribe messaging protocol specifically designed for IoT environments with limited bandwidth, high latency, or intermittent connections. It uses a broker to mediate messages between clients and supports three Quality of Service (QoS) levels, allowing developers to balance between performance and reliability. MQTT is widely adopted in scenarios such as telemetry, smart home automation, and device monitoring. However, its reliance on a centralized broker and the TCP protocol can add complexity and overhead, particularly in extremely constrained or decentralized systems.

CoAP. CoAP (Constrained Application Protocol), on the other hand, is a web transfer protocol designed for constrained devices and networks. It operates over UDP and follows a RESTful model similar to HTTP, making it easy to integrate with web-based services. CoAP supports confirmable and non-confirmable messages, enabling optional reliability while retaining the lightweight benefits of UDP.

Additionally, it allows for multicast communication, which is advantageous in sensor networks or when broadcasting updates to multiple devices. Security in CoAP is typically handled via DTLS (Datagram Transport Layer Security). While CoAP is efficient and tailored for constrained environments, its ecosystem is not as mature or widespread as MQTT's, and the reliance on UDP means developers must handle certain aspects of reliability themselves if needed.

IoT protocols at application layer

MQTT. MQTT (Message Queuing Telemetry Transport) is a lightweight, publish/subscribe messaging protocol optimized for low-bandwidth, high-latency, or unreliable networks. Originally developed by IBM, MQTT uses a broker to mediate communication between clients, making it ideal for scenarios like remote monitoring, smart homes, and telemetry systems. It supports three Quality of Service (QoS) levels to manage message delivery guarantees, which is particularly useful in IoT environments where devices may frequently go offline. MQTT is popular for its low overhead and efficient message distribution, although it relies on a centralized broker and operates over TCP, which may introduce some complexity and overhead in highly constrained settings.

XMPP. XMPP (Extensible Messaging and Presence Protocol), originally designed for instant messaging, is also used in IoT scenarios that require real-time communication, presence information, and extensibility. XMPP supports both publish/subscribe and peer-to-peer models and is known for its flexibility and use of XML for data formatting. While it's not as lightweight as MQTT or CoAP, it is useful in certain use cases like smart grids, connected vehicles, or collaborative applications where continuous communication is important.

HTTP/HTTPS. HTTP (HyperText Transfer Protocol) and its secure version, HTTPS, are widely used in IoT, particularly for device-cloud communication and RESTful APIs. HTTP is easy to implement, compatible with existing web infrastructure, and well-understood by developers. However, it is relatively heavy and not ideal for battery-powered or resource-constrained devices due to its verbose headers and reliance on TCP. Still, in many commercial and consumer-grade IoT systems—especially those involving firmware updates, configuration interfaces, or integration with web platforms—HTTP/HTTPS remains a practical choice.

AMQP. AMQP (Advanced Message Queuing Protocol) is a robust, enterprise-grade messaging protocol designed for message-oriented middleware. It provides advanced features such as message queuing, routing, reliability, and security, making it ideal for industrial IoT and large-scale deployments. However, its complexity and resource requirements make it less suitable for constrained devices. AMQP is commonly used in back-end systems where guaranteed delivery and high reliability are essential, such as in logistics, finance, or manufacturing automation.